Impedance-controlled manipulator with Series-Elastic Actuators

Group 04

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1 Introduction

The model was built with the goal of keeping it as realistic as reasonably possible in most aspects, such as arm's and motors' parameters, control, friction modelling and so on. The objective of the model is to track a setpoint moving in a circular or butterfly-like trajectory with a frequency less than or equal to 2 $\mathrm{rad\,s^{-1}}$ with reasonable precision, while also keeping it safe to interact with.

There were two critical design choices for the robot arm. First was the choice to use the Maxon motor and gearbox combination, this was done due to the familiarity of the components and their behaviour in simulation, the main effect it had was a limitation on the size of the robot since the maximum output torque for each actuator is 3.4 N m. The second choice was to make the robot links aligned, this is conventionally avoided to prevent self collisions and allow each joint to be used to its full range of motion; however, a collision avoidance controller was also chosen to be developed, so to facilitate its testing self collisions were facilitated.

2 Model

The robot arm is made from 3 rotational joints in a yaw, roll, roll configuration (ZXX) and it's "home" or zero position such that the first link in pointing up and the other two are alligned to the Y axis.

The arm was modelled in the 3D-Mechanics toolbox assuming it to be made out of aluminium tubes, with a density of $\rho = 2700\ \mathrm{kg\,m^{-2}}$, with outer ($r_{ext}$) and inner ($r_{int}$) diameters of 40 and 35 mm respectively. The lengths of the links were chosen in such a way that the selected motors could counteract gravity in the stretched out configuration while being of multiple of 10 cm length. To geometrically prevent the arm from damaging it's own base, links 1 and 3 were set to be of the same length, and link 2 to be twice as long. The values were chosen as 0.1 (l_1), 0.2 (l_2) and 0.1 (l_3) meters. The motor-gearbox assemblies were assumed to be solid cylinders with dimensions in accordance with the datasheets and total masses rounded up to 0.5 kg to account for coupling elements, cables, or any other unmodeled weight.

Using this data, links' masses are computed using

$$\pi \cdot (r_{ext}^2 - r_{int}^2) \cdot l_i \cdot \rho,\ i = 1, 2, 3. \tag{1}$$

The inertias about x (I_{xx}) are computed the same way for each link using

$$\pi \cdot l_i \cdot \rho \frac{r_{ext}^4 - r_{int}^4}{4} + l_i^2 \frac{r_{ext}^2 - r_{int}^2}{12},\ i = 1, 2, 3. \tag{2}$$

About y (I_{yy}), the inertia for link 1 is computed using

$$\pi \cdot l_1 \cdot \rho \frac{r_{ext}^4 - r_{int}^4}{4} + l_1^2 \frac{r_{ext}^2 - r_{int}^2}{12}, \tag{3}$$

while for link 2 and 3 this is

$$\pi \cdot l_i \cdot \rho \frac{r_{ext}^4 - r_{int}^4}{2}, \ i = 2, 3. \tag{4}$$

Finally for the inertias about z (I_{zz}), these equations are swapped around, so for link 1 it is

$$\pi \cdot l_1 \cdot \rho \frac{r_{ext}^4 - r_{int}^4}{2}, \tag{5}$$

and for links 2 and 3 it becomes

$$\pi \cdot l_i \cdot \rho \frac{r_{ext}^4 - r_{int}^4}{4} + l_i^2 \frac{r_{ext}^2 - r_{int}^2}{12}, \ i = 2, 3. \tag{6}$$

The results of these calculations are shown by Table 1. Figure 1 shows the 3D model.

Table 1: Mass and inertia of each link.

	Mass [kg]	I_{xx} [kg m^2]	I_{yy} [kg m^2]	I_{zz} [kg m^2]
Link 1	0.147	2.27×10^{-4}	2.27×10^{-4}	2.08×10^{-4}
Link 2	0.295	119×10^{-3}	4.16×10^{-4}	119×10^{-3}
Link 3	0.147	2.27×10^{-4}	2.08×10^{-4}	2.27×10^{-4}

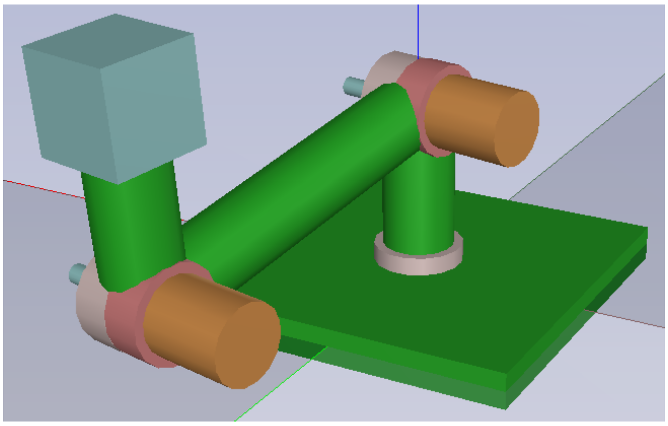


Figure 1: 3D model of the ZXX manipulator.

2.1 Theoretical background

2.1.1 Controllers gains choice

The gains choice in the impedance and torque controllers is mostly based on techniques presented in [1], although the main novelty of the paper, namely elastic element stiffness optimization, turned out to be very difficult to apply to Cartesian space impedance controller. After a failed attempt to bring the paper's results into the Cartesian space, it was decided to tune the elastic elements' stiffness manually and use the insights on transparency and torque tracking properties to validate the correctness of the choice.

The impedance controller gains were calculated by ignoring the Coriolis and gravity terms (assuming low joint velocity and perfect gravity compensation) and representing the system in the second order form:

$$\Lambda \ddot{x_{ee}} = K(x_{des} - x_{ee}) + b(\dot{x}_{des} - \dot{x}_{ee}),$$

where x_{ee} is the end-effector x coordinate, x_{des} is desired x position, K and b are damping and stiffness of the impedance controller and Λ is the effective mass of the robot at the end effector in x direction. x

can be replaced to any other direction or even to end-effector's Cartesian coordinates, but since damping and stiffness in impedance controller are usually set in x, y and z directions, and in our case it would make sense to set them all equal, this assumption provides a simpler calculation. Additionally, assuming x_{des} and $\dot{x}_{des}$ to be 0, we notice that the equation becomes the canonical second order system equation:

$$\Lambda \ddot{x}_{ee} + b\dot{x}_{ee} + Kx_{ee} = 0,$$

$$K = \omega_n^2 \Lambda \quad b = 2\Lambda\omega_n\xi.$$

Since the objective is to track trajectory of frequency 2 rad/s, to have some margin we set system's natural frequency $\omega_n = 4$ rad/s. There is no need for a fast step response, thus we set $\xi = 2$ to improve tracking. Λ depends on the robot's configuration. Assuming joints 2 and 3 are in such position that the arm is stretched along y axis, and its moment of inertia about z is around $0.05\,\text{kgm}^2$. To get the effective inertia along x, it is divided by the squared distance from z axis, which is the sum of lengths of links 2 and 3, resulting in $\Lambda \approx 0.55$ kg. Finally, we get $K \approx 9$ N m and $b \approx 9$ Ns/m, which seem reasonably low to provide safe interaction, but in a non-educational scenario should be confirmed with regards to specific use-case scenarios.

The choice of torque control gains follows the same procedure as in [1], but the damping terms are disregarded since they are not present in our model. It starts with inspecting the relation of the elastic element deformation to the desired torque and load displacement, where the denominators have a form of a 2-nd order system:

$$\Delta_{\tau^*}(s) = \frac{\Delta(s)}{\tau^*(s)} = \frac{K_d s + K_p + \lambda(s)}{Is^2 + D_\Delta s + K_\Delta}, \quad \Delta_{ql}(s) = \frac{\Delta(s)}{q_l(s)} = \frac{-Is^2}{Is^2 + D_\Delta s + K_\Delta},$$

where

$$D_\Delta = k_b K_d, \quad K_\Delta = k_b(K_p + 1),$$

and I is the combined inertia of the motor and the gearbox, K_p and K_d are the gains of a PD controller. k_b is the elastic element stiffness, which was empirically tuned to achieve precise enough tracking and reasonable transparency and torque tracking properties, more on that in section 3. Assuming locked-output (or a load inertia being much higher than rotor inertia), from there the following relation for pole placement emerge:

$$K_d = \frac{2I\omega_0\zeta}{k_b}, \quad K_p = \frac{\omega_0^2}{\omega_n^2} - 1,$$

where $\omega_n = \sqrt{k_b/I}$ is the natural frequency of the SEA. Based on it ω_n can be set as $\omega_n = \alpha\omega_0$ leading to $K_p = \alpha^2 - 1$. Having a very low inertia of $I = 3.55 \cdot 10^{-6}$, the natural frequency $\omega_n \approx 1680$ Hz is significantly higher than the frequency of the target trajectory. Thus, no need for high α. It was set to $\alpha = 3.5$ because with lower values the tracking performance was considerably worse. To make the system critically damp we set $\xi = 1$, resulting in $K_d \approx 0.004$.

Although the arm performed the tracking tasks well, we observed that during gravity compensation tests, when the only control input is the $\mathtt{g}$ term, it does not float but moves and occasionally accelerates. After double-checking everything and trying different control gains, we attempted to replace the PD controller in the torque loop with PID, which did improve the situation. The I gain was manually tuned, with the final value of 60. This change does make the second order system analysis invalid, but the selection of α and ξ is strongly based on the simulation results anyway.

2.1.2 Collision avoidance

For the collision avoidance controller the concept was inspired by [2] but developed to a much simpler level and adapted to fit in conjunction to the torque control. In essence, spheres are placed along the robot's links, their radius and a safety distance between them is defined, if the safety distance is violated a force with direction opposite to the distance between the spheres centre's is calculated based on a desired stiffness and finally it is projected into the joint via the Jacobian of the sphere in its link. Using a similar approach but instead of comparing two spheres, comparing a sphere against the ground plane, collisions with the ground are also avoided.

The repulsive force is calculated as

$$\mathbf{F}_{ij} = K_c \left(d_s - \|\mathbf{p}_i - \mathbf{p}_j\|\right) \hat{\mathbf{n}}_{ij}, \qquad \|\mathbf{p}_i - \mathbf{p}_j\| < d_s, \tag{7}$$

where K_c is the self collision stiffness, d_s is the safety distance, p_i and p_j are sphere centres and $\hat{\mathbf{n}}_{ij}$ is the normalized vector between sphere centres.

For ground plane avoidance K_c is replaced by K_g, which is the stiffness against collisions with the ground and $\mathbf{p}_j$ can be replaced with $[\mathbf{x}_i, \mathbf{y}_i, 0]^T$.

The torque in each link is obtained as follows,

$$\boldsymbol{\tau}_{\text{self}} = \sum_{(i,j)\in\mathcal{N}_f} \mathbf{J}_{s_i}^T \mathbf{F}_{ij} \tag{8}$$

, where i and j are the indices of two different spheres, N_s is the number of spheres and $i \neq j$ since a sphere cannot, or is always in collision with itself.

Special attention is brought to the way the Jacobians are computed for the spheres, which relates to the way their positions are defined. Instead of computing all of them completely analytically, only the Jacobians for each joint are calculated in detail and the Jacobians for each sphere are interpolated from the joints. The position of a sphere along a link can be defined as

$$P_s = P_{i-1} + \alpha(P_i - P_{i-1}), \qquad 0 \leq \alpha \leq 1, \tag{9}$$

where P_i is the position of the next joint and P_{i-1} is the position of the joint that actuates the current link, and α is the relative position of a sphere along a link's length.

Similarly to how the position of a sphere can be defined in this way, the Jacobian of a sphere can be interpolated in the same way as

$$J_s = J_{p_{i-1}} + \alpha(J_{p_i} - J_{p_{i-1}}), \qquad 0 \leq \alpha \leq 1, \tag{10}$$

where J_{p_i} is the Jacobian until the next joint and $J_{p_{i-1}}$ is the Jacobian until the joint that actuates the current link, and α is the relative position of a sphere along a link's length. In this way sphere positions can be defined with a single parameter and the adjacent "partial" Jacobians.

2.2 Model overview

The model implements an impedance control loop with SEA at each joint. The overall system is shown in the Figure 2. The control structure is shown in Figure 3. The impedance controller operates at the highest level, then the torque control for the SEA in each joint, and at the lowest level the current controller for each motor, the plant is the robot arm model implemented in the 3D Mechanics. An additional block of Coll_avoid implements collision avoidance in parallel with the impedance controller. Other features are references that define the trajectory for the impedance controller and external forces acting on the end effector. In block Control_gains all control parameters are calculated. The model is almost fully parametrised - a change in arm's physical parameters automatically reflects on the control gains and trajectory definition.

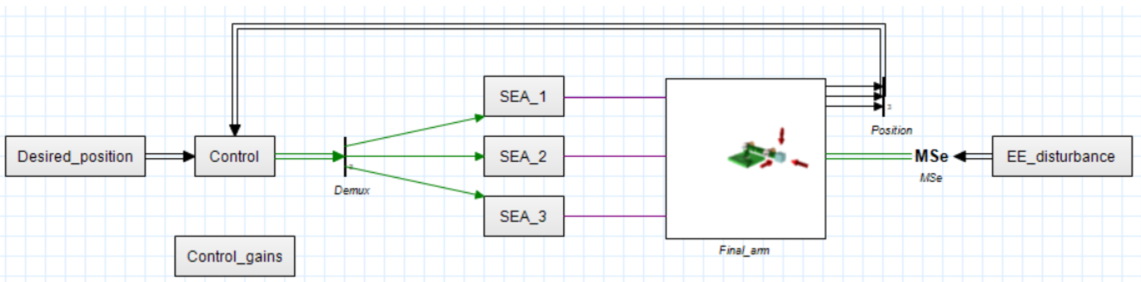


Figure 2: Full Model

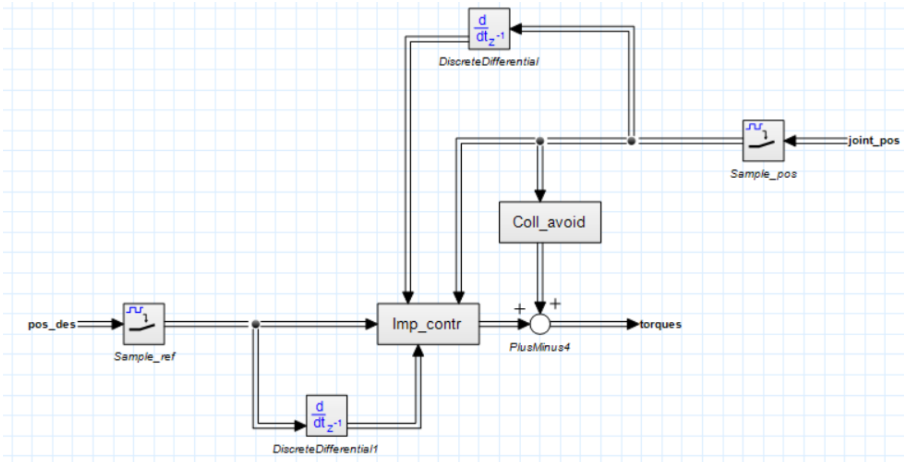


Figure 3: Control block

The impedance controller is the core of the whole control system. The model is written in an equation submodel. There are 4 input signals; reference trajectory, desired velocity of the end-effector, the robot encoder measurements of angular position, and angular velocity of each joint. Velocities are obtained via DiscreteDifferential blocks. In the case of encoders, this is a realistic scenario, but for the desired velocity, they could also be obtained directly from a user-defined source or a more complex trajectory generator; however, it was considered to be out of scope of the assignment.

The SEA block is shown in Figure 4. The block takes the desired torque at each joint and outputs the actuation torque at each joint of the robot model. The Maxon Motor model inside the SEA block is shown in Figure 5. This block takes as input the reference current and outputs the torque at the rotor connected to the elastic element. These two submodels compose the SEA with cascaded control of an inner-loop PI controller for current, and an outer-loop PID + Feedforward controller for torque.

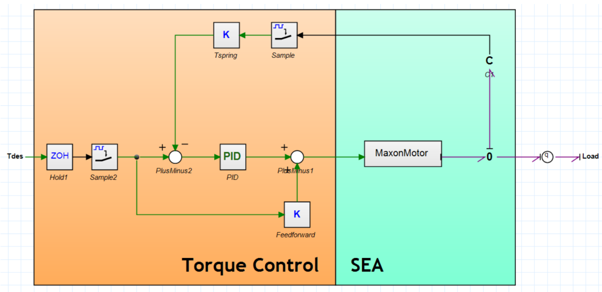


Figure 4: SEA at each joint

The inner current controller regulates the motor current in the H-bridge driver, and the regulated current is converted into the mechanical torque that moves the rotor. The block takes into account the conversion from the electrical domain into the mechanical domain, as well as power loss in the gearbox due to friction. The PI controller is chosen because the electrical dynamics consists of the first order differential equation with the inductance and the resistance. In addition, the robot arm performs in relatively low speed so the backEMF effect is neglectable, therefore the PI controller is sufficient in this application.

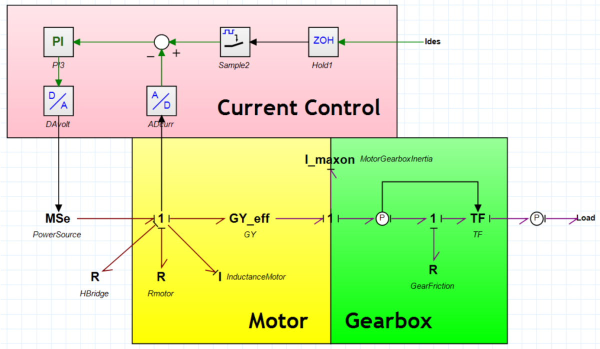


Figure 5: Maxon Motor Current Control

Finally, the different controllers run at different frequencies, which are allowed by the Sample and ZOH Blocks. The impedance and torque controllers run at 1 kHz and the current controller which is at the lowest level runs at 10 kHz.

2.3 Modelling choices

For the motor model, the main modelling choice was to implement a modified gyrator that accounts for motor efficiency. This is performed via a variable of dynamic efficiency, which checks the direction of power flow and ensures less power flows out than what came in regardless of which port it comes from. Related to efficiency, the motor is modeled not only as the gyrator but also with the inductance and resistance in series with values according to the datasheet. The R element accounts for the electrical power losses, which are already considered in the 88% used in the modified gyrator, so the efficiency in the model will never reach the reported maximum efficiency because these losses are accounted for twice. The effect of this redundancy is mostly present in non representative input voltages, however, the effect in simulation for the robot arm is minimal.

The gearbox was modeled as the combination of a transformer and an R element to represent friction, in this way, friction accounts for the efficiency of the gearbox instead of making it a plain parameter as in the motor. The transformer changes speed exactly according to the gear ratio, assuming always rigid and constant contact. The friction itself is modeled both with Coulomb and viscous friction, ignoring the Stribeck effect, and represented with a smooth function as,

$$F_f = \tau_c tanh(kv) + bv \tag{11}$$

,where τ_c is the Coulomb friction coefficient, k is the smoothness factor, b is the viscous friction coefficient and v is speed.

To determine these parameter values a relatively artisanal method was performed, first the smoothing factor was found to be around 50-200 for metal on metal contact and from testing the lower end seemed more accurate. Then the inbuilt 20-sim optimization was used to obtain the two coefficients such that the efficiency was about 75% as stated in the datasheet, and the coefficients were somewhat balanced and within the same order of magnitude so as to ensure both friction types were represented realistically. The trajectory for the optimization was an oscillation because it constantly crosses the static boundary and also reaches considerable speed so both types of friction phenomena occur in balance. Due to this procedure, the gearbox efficiency during a test at minimal speed like only gravity compensation can be much lower than 75% and with constant torque input it is higher, but generally provides an accurate representation.

3 Simulation

3.1 Verification of the Model

To verify if the model behaves correctly, several tests have been done. As a first test, each motor's MSe input was set to 0 V. The result of this is shown in Figure 6. Here, it can be observed that the arm falls down from its starting position without any oscillations. This is expected, because even though the motor voltage is set to 0 V, the arm still backdrives the motor and it therefore generates a back-EMF of $v_{emf} = k_e \cdot \omega$. This creates a viscous braking torque at each motor and thus slows down the fall and damps oscillations.

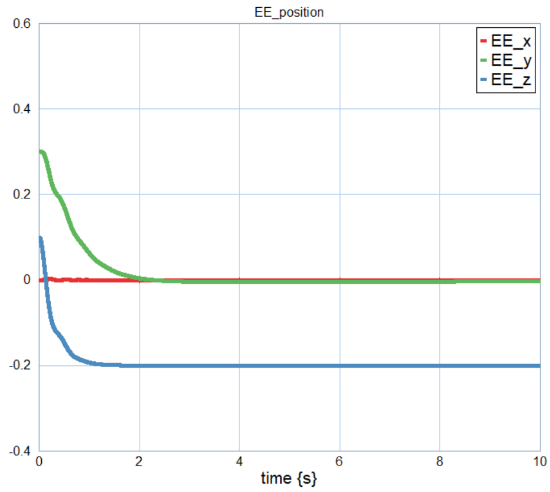


Figure 6: Motor voltage set to 0 V, back-EMF causes damping [`G04_BackEMF.emx`].
Video

To see the arm's behaviour with zero actuation, the current in the motor's H-bridge was forced to 0 A using an `Sf` element. Figure 7 shows the results of this. As expected, with absolutely no actuation coming from the motors, the arm oscillates before falling down to its stable equilibrium, just like a double pendulum. It can also be seen that there are slight oscillations along the x-axis. This is can be explained by the motors being mounted on the side of the robot. Since these motors masses are off-plane, and are modelled as such, this causes a yaw rotation through inertial coupling.

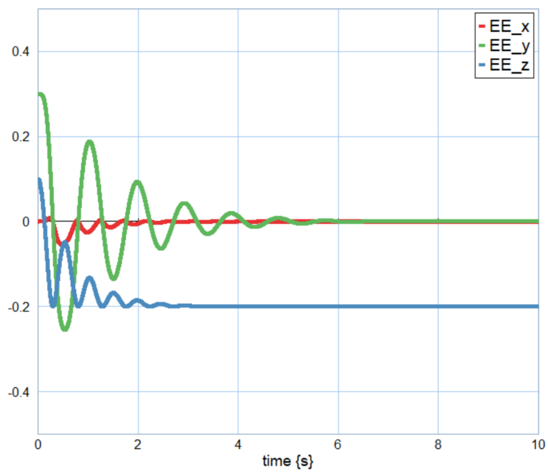


Figure 7: Motor H-bridge current set to 0 A [`G04_NoActuation.emx`].
Video

The next test is the gravity compensation in isolation. This means that the control law is simply $u = g$, impedance control is left out. These results are shown by Figure 8. Here, it can be seen that the arm “sags” a bit. This can be explained by the fact that, while the tension in the spring is building up, the link has some time to start moving. By the time the tension in the spring is large enough, the link will keep moving at a constant speed because all the forces are balanced.

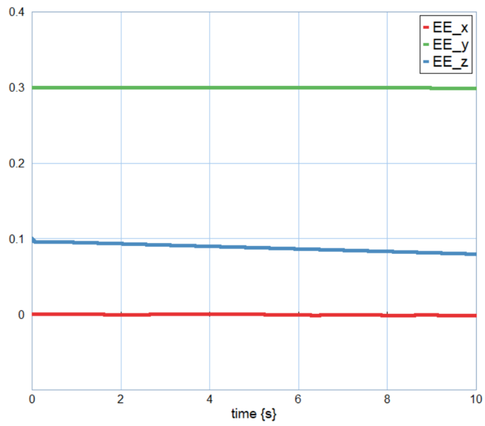


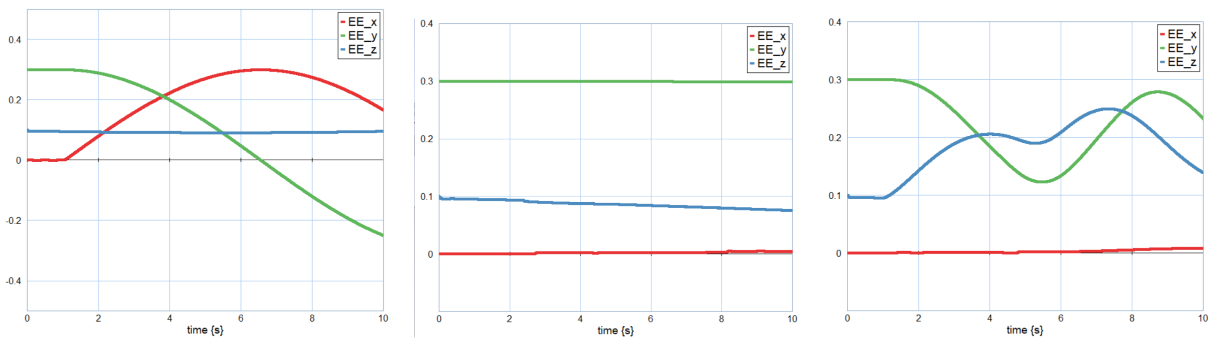
Figure 8: Gravity compensation only[G04_GravityComp.emx].
Video

For the final verification step, the gravity compensation was tested alongside a disturbance along each axis. Figure 9 shows three separate plots of how the system reacts to disturbances in all three directions.

Figure 9a shows a disturbance in the positive x direction which, as expected, results in a yaw rotation about the z-axis with x initially increasing.

Figure 9b then shows a disturbance in the positive y direction. Again, the arm behaves as expected. Since this disturbance is in plane of the robot’s orientation, its behaviour does not change much from the gravity compensation test. It can be seen that the arm straightens out slightly in the y-direction, meaning the y coordinate does not decrease as much.

Finally, Figure 9c shows a disturbance in the positive z direction, or upwards. It can be seen that the end effector does indeed start moving upwards, but then gets stopped and reverses directions. This reversal is due to the collision avoidance stopping the end effector from going into the second link, thus supplying a force in the opposite direction in joint 3. The end effector can be seen going back and forth a few times like this, but ultimately this is expected behaviour.



(a) Disturbance in the positive x direction of magnitude 0.5 for 0.1 seconds [G04_DisturbanceX.emx].
Video

(b) Disturbance in the positive y direction of magnitude 0.5 for 0.1 seconds [G04_DisturbanceY.emx].
Video

(c) Disturbance in the positive z direction of magnitude 0.1 for 0.1 seconds [G04_DisturbanceZ.emx].
Video

Figure 9: Three plots showing gravity compensation, along with a disturbance in x, y, and z respectively.

3.2 Validation of the Model

As discussed above, since the elastic element stiffness was chosen manually based on the model performance, the correctness of this choice was validated by inspecting the emerging torque tracking and transparency properties of the end effector. For conciseness, only x direction is analysed, thus only the first joint contributes to the measurement. Additionally, the usual stretched-out configuration in which the arm was analysed above, was replaced with a slightly bent one, setting initial position of joint 2 to $-10°$, and joint 3 to $20°$. The reason for this change is that the integration process for these experiments fails in the singular configuration.

Torque tracking characterises how well an actuator can apply a periodic torque in the case of fixed output. In our case though, where the output is the end-effector of a 3-DoF arm, where output is its end-effector, it would make sense to measure how well it can apply a periodic force, rather than torque. To keep the naming in accordance with [1], we will continue to use the term "torque tracking". In the validation setup a source of effort with a constraint function is attached to the disturbance port of the 3D mechanics model to fix it in place. The torque input to the SEAs is calculated by applying a transpose of the Jacobian to the desired force. The desired force in x is a sweep block that generates a sine signal with gradually increasing frequency from 0 to 50 Hz. The plot of the desired force and the reaction force at the end-effector is shown in Figure 10. In the case of perfect torque tracking, the two lines would be symmetric over the time axis, but we observe that the actual applied force is at first higher than the target (probably due the to transient component of the dynamics), and it starts decreasing around 4 seconds or 20 Hz.

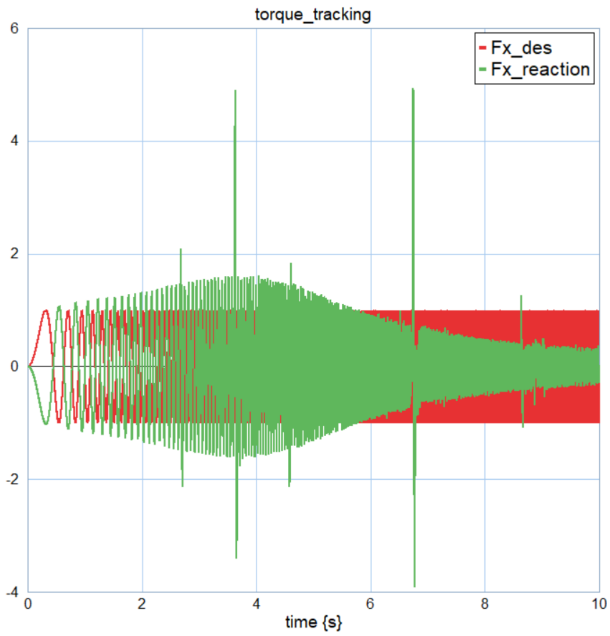
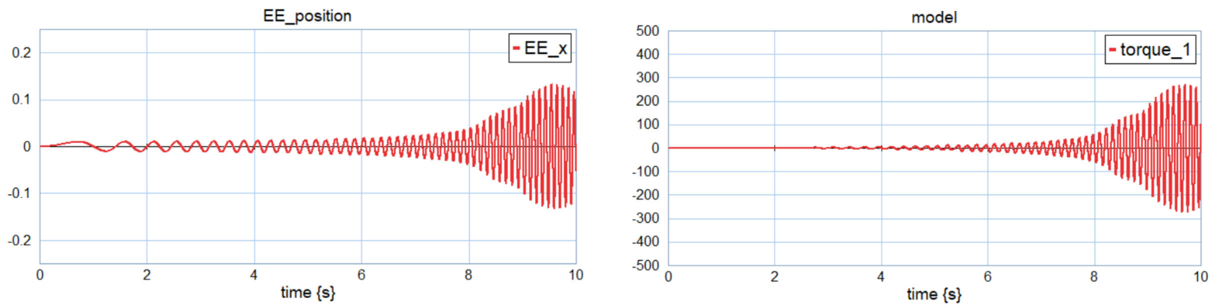


Figure 10: Desired force and actual force at the end-effector in x direction with increasing frequency[`G04_TorqueTracking.emx`].
Video

Transparency is the ability of an actuator to provide zero torque in the case of a moving load due to an external force. It was attempted to inspect it in the end-effector frame as well. Although the setup for the transparency inspection is not as clean as that for torque tracking. When it was attempted to constrain the movement of the end-effector to a certain variable velocity in a similar fashion, the program always raised "BDF is not making good progress" error in the middle of a simulation, regardless of the arm configuration and velocity amplitude. In the final implementation, the motion of the end-effector is governed by a modulated source of effort controlled by 3 PID controllers, one for each axis, whose gains were initially derived in the same way as for the impedance controller and then tuned manually. This setup does not allow to set the trajectory precisely, especially for high frequencies, but is enough to get the general idea of the transparency limits. The desired trajectory is again defined by a sweep function that starts at 0 Hz and approaches 10 Hz. The setpoint torque for each SEA is set to 0. We plot the torque at the first joint and not the force felt at the end-effector in the x direction because it consists of both the force induced by the SEA and the arm's dynamics. The result is shown in Figure 11. In the case of ideal transparency the torque in joint 1 would always be zero, but we notice that it starts growing at around 4 seconds, which corresponds to 4 Hz.



(a) End-effector position moved under an external force

(b) Torque in joint 1 with 0 setpoint

Figure 11: Transparency test[`G04_Transparency.emx`].
Video

Therefore, we observed that in this configuration, the chosen spring stiffness and torque control parameters allow the arm to accurately apply a periodic force of up to 20 Hz or 126 rad/s, which is considerably higher than the frequency of the chosen target trajectory of 2 rad/s. The transparency characteristic should be compared with the safety requirements, which we did not define in any way. However, the actuator is able to provide almost zero torque up to a frequency of 4 Hz, which can be considered reasonable to achieve safe interaction with a human.

To validate the functionality of ground collision avoidance the system was tested along with gravity compensation, such that as it fell and approached the ground it would be repelled, the result is shown in Figure 12. As expected, the robot falls slowly, and as the end effector reaches the 0.04 cm threshold from the ground, torques are applied to prevent a crash and it switches direction.

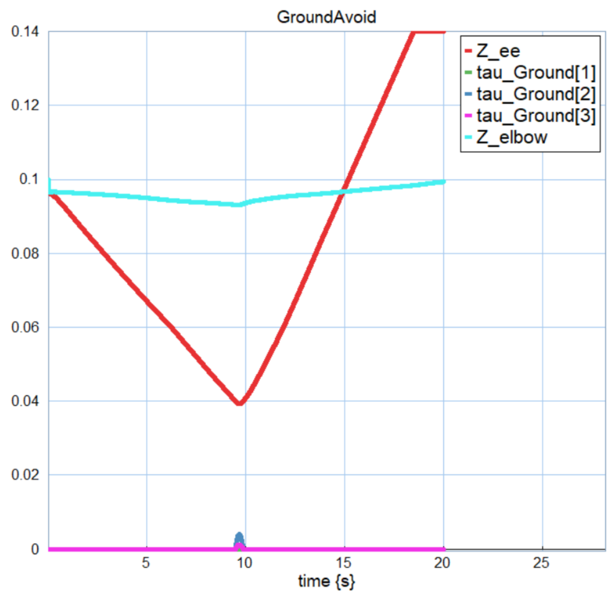


Figure 12: Ground Avoidance test[`G04_GroundAvoid.emx`].
Video

The functionality of the impedance controller was tested without collision avoidance to evaluate tracking performance and resistance to disturbances, the result is shown in Figure 13. The test follows a circular trajectory in the XZ plane and a disturbance force of 2 N in Y occurs from 4.0 to 6.0 seconds. The behavior is as expected, first a transient phase where the robot moves from its original position to the reference, then it follows it closely, after that it is complies with the disturbance and finally it returns to the desired trajectory.

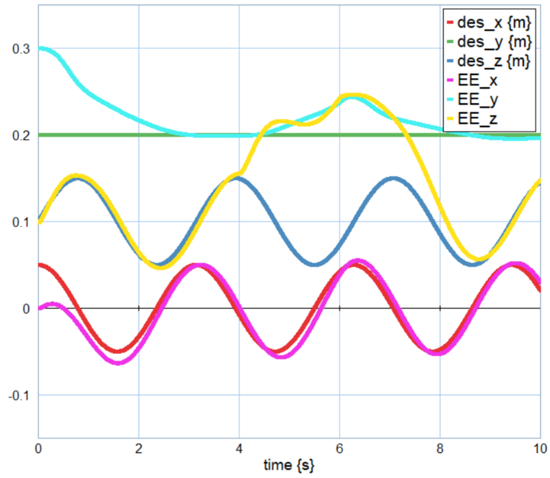


Figure 13: Impedance control and disturbance test[`G04_ImpContr.emx`].
Video

Finally, the full impedance and collision avoidance was tested, the trajectory is the same as the previous test but the crashes with the ground are prevented, the result is shown in Figure 14. The expected behavior is that the robot follows the trajectory until the elbow comes close to the ground at which point this joint is not allowed to go further down and there is a recurring error in Y and Z. The observed behavior is different because the stiffness for ground collision is not that high, because this presented problems in other cases, so the robot goes further than it should but still presents a clear opposition to crashing with the ground. The robot could technically perform this trajectory with the elbow pointing up instead of down, avoiding this issue, but the mirroring configuration is chosen depending on initial configuration and disturbances.

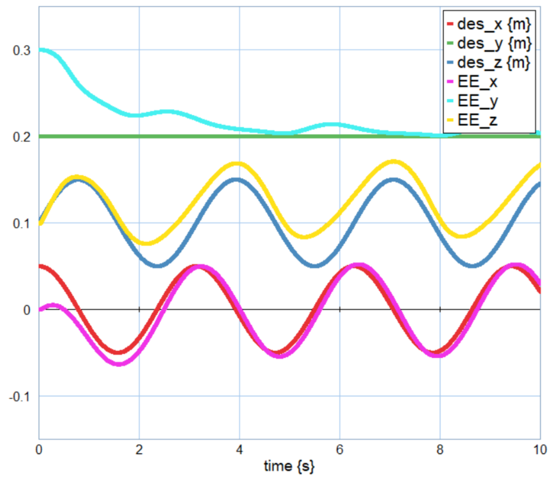


Figure 14: Impedance control and collision avoidance [`G04_ImpContrCollAvoid.emx`].
Video

4 Discussion & Conclusion

The developed model could serve as a representation of an actual robotic arm. It's mechanical properties were derived from certain assumptions on the materials used and can easily be adjusted through parametrization. Mechanical and electrical domains of the SEAs were modelled with enough rigour to serve as a good representation of reality, taking into consideration efficiency, Coulomb and viscous friction and higher sampling rate for the current control loops. Impedance and torque control gains were calculated based on the plant properties, although some guesswork was used. We still don't have a proper explanation to why the gravity compensation did not work well until we replaced the PD controller with PID in the torque control loop, but we theorize the steady state error may be due to discretization. Eventually, the arm started to behave as expected, which was demonstrated in the verification section. Since the stiffness of the elastic element was chosen based on the observed quality of trajectory following, we explored the torque tracking and transparency properties of the end-effector, which emerge from it interacting with torque control loop and arm's dynamics. The obtained results appear to be reasonable for some possible use cases. Finally, obstacle avoidance algorithm was developed and tested, demonstrating expected behaviour.

During the completion of this assignment, we improved our understanding of the transparency and torque tracking properties of an SEA and learned a method to implement obstacle avoidance. Regarding the model, we learned that it's triple cascaded control loop is more sensitive to a change in gains than we expected. During gravity compensation verification, we observed that a very small change can lead to drastically different behaviour. We also got the chance to try some additional content in the form of the collision avoidance and were amazed by the simplicity by which it can be implemented by harnessing our prior knowledge of Jacobians.

References

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